DELUXE STORE
DATA PREPARATION DOCUMENTATION

Introduction

This document describes the data preparation process for the Deluxe Store capstone project. The purpose is to assess the quality of the sales dataset, identify potential data quality issues, document the cleaning decisions made, and prepare the data for accurate business analysis and visualization.

Dataset Overview

The dataset used for this project is the Deluxe Store Superstore sales dataset. It contains sales transactions and related information about orders, customers, products, locations, shipping, discounts, and profit. The dataset will be examined and prepared before being imported into Power BI for analysis and visualization.

Data Quality Log

Check	Intial result		Decision
Total Rows		9,994	Record baseline
Total Columns		24	Retain relevant fields
Missing Values	None detected		No missing-value treatment currently required
Exact Duplicate Rows	1 duplicate occurrence		Verify and remove one copy in prepared data
Negative-Profit Records		1871	Retain for analysis
Data Fields	Order Date and Ship Date present		verify data types and consistency

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1. Data Quality Assessment

The dataset was reviewed to identify potential data quality issues before analysis. The checks included duplicate records, missing values, data types, and unusual numerical values.

2. Duplicate Records

During the duplicate check, two identical records were identified. One copy was removed from the working dataset to prevent the same transaction record from being counted twice. The original dataset was retained unchanged.

3. Data Quality Checks

Missing Values Check

- Check performed: Checked the dataset for blank cells.
- Initial result: No blank cells were identified during the initial check.
- Decision: No changes were required for missing values at this stage

Data Type and Value Checks

Date Fields: The Order Date and Ship Date columns were formatted as short dates to ensure consistent date presentation.

Sales and Profit: The Sales and Profit columns were formatted to display two decimal places.

Quantity: The Quantity column was checked, and no zero or negative values were found.

Discount: The Discount column was checked. No values above 1 (100%) were found. Zero discounts were retained because they indicate transactions without discounts.

Profit: Negative profit values were retained because they represent transactions where the store incurred losses.

4. Final Dataset Preparation

After removing one exact duplicate record, the prepared dataset contained 9,993 rows and 24 columns. The original dataset contained 9,994 rows and 24 columns.

The negative-profit records were retained because they represent genuine business losses and are important for profitability analysis. Records with zero discounts were also retained because they represent transactions without discounts.

The prepared dataset was saved as an Excel workbook and imported into Power BI for further analysis and dashboard development.

5**. Conclusion**

The dataset was reviewed, cleaned, and validated, resulting in a prepared dataset of 9,993 rows and 24 columns. The prepared dataset was then used for further analysis and dashboard development in Microsoft Power BI.